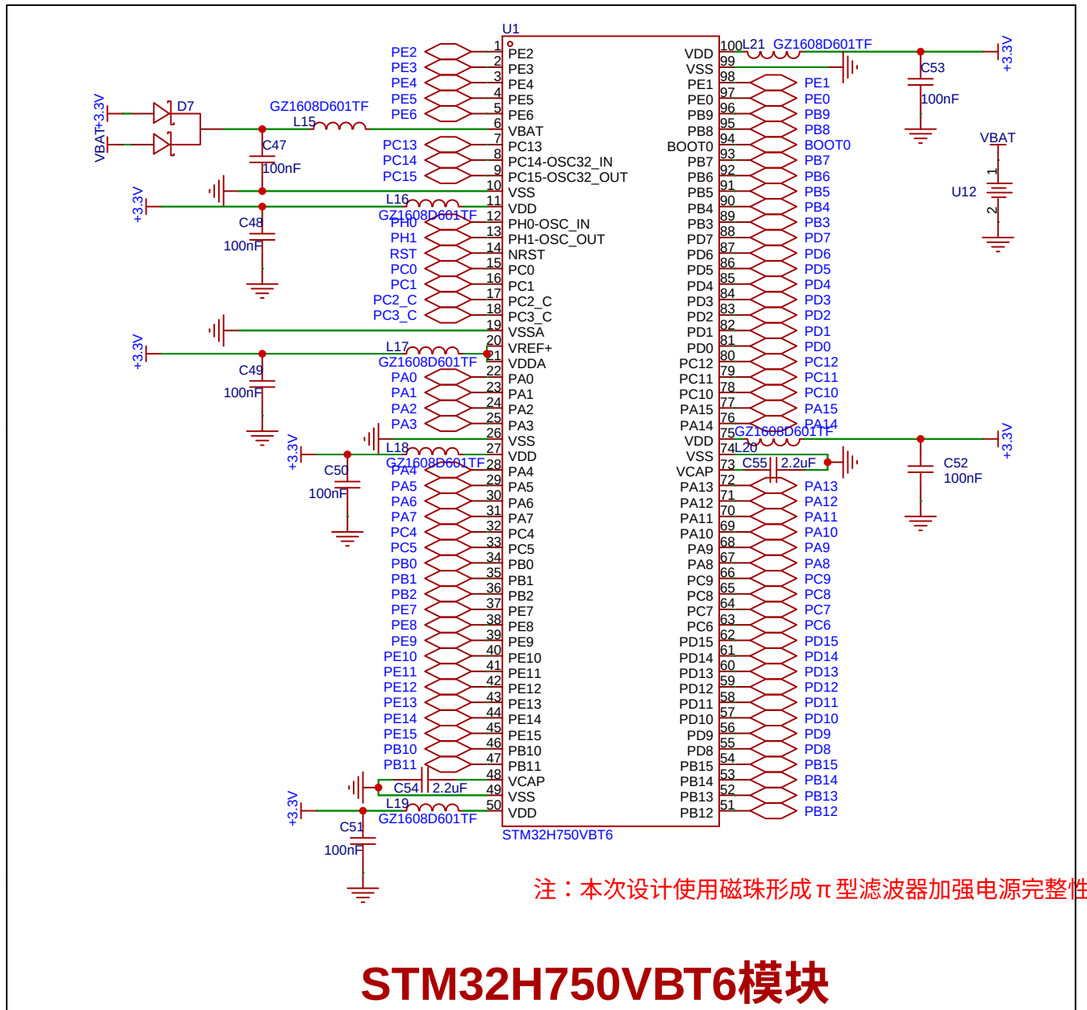
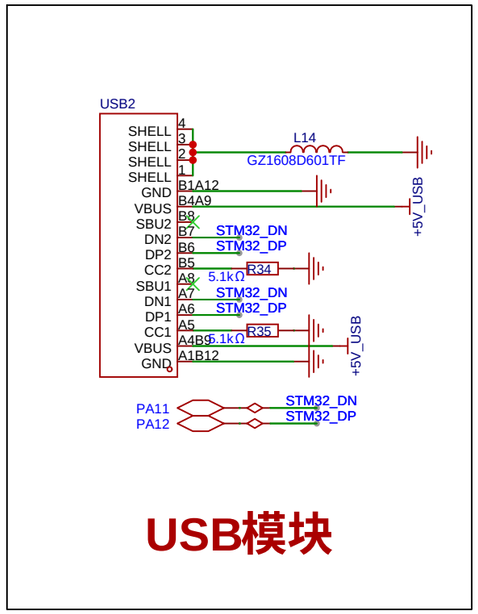
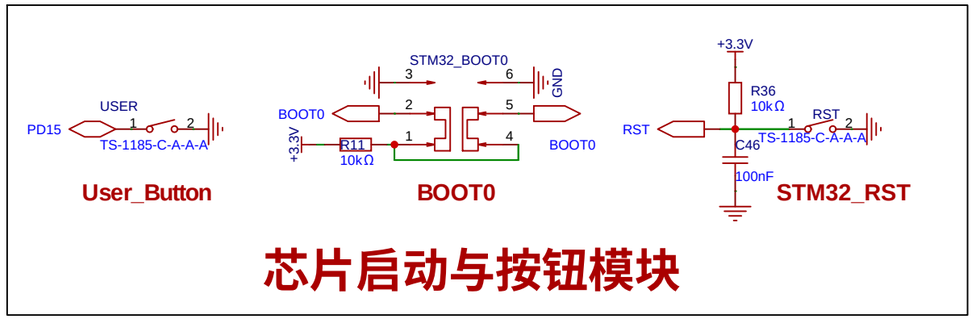


注：本次设计使用磁珠形成π型滤波器加强电源完整性

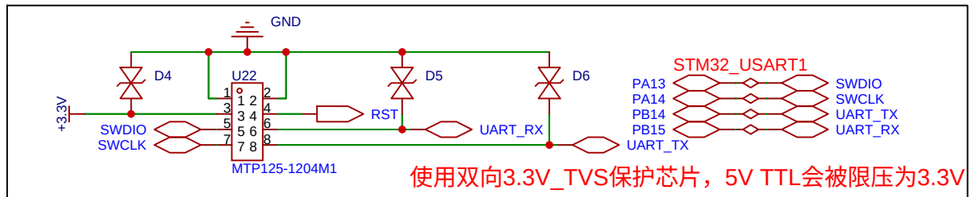
STM32H750VBT6模块



USB模块

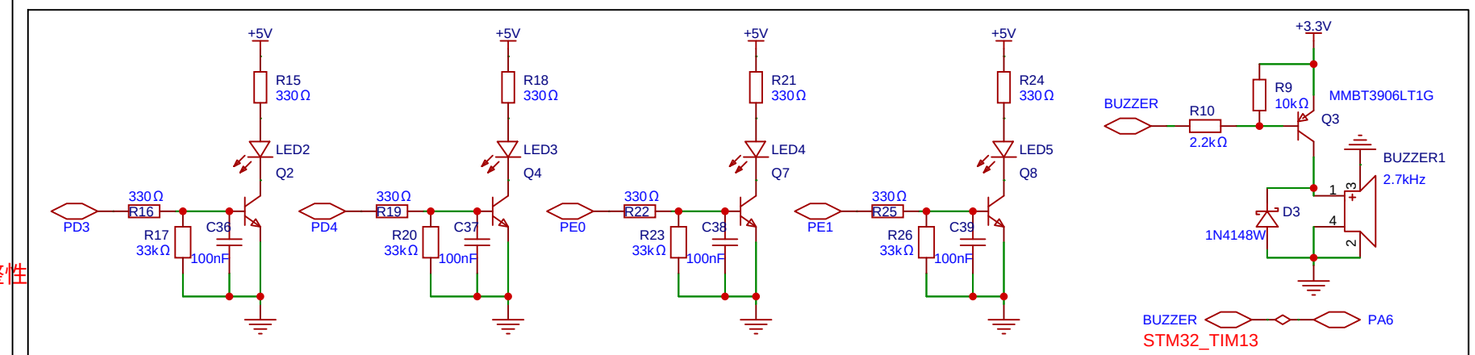


芯片启动与按钮模块

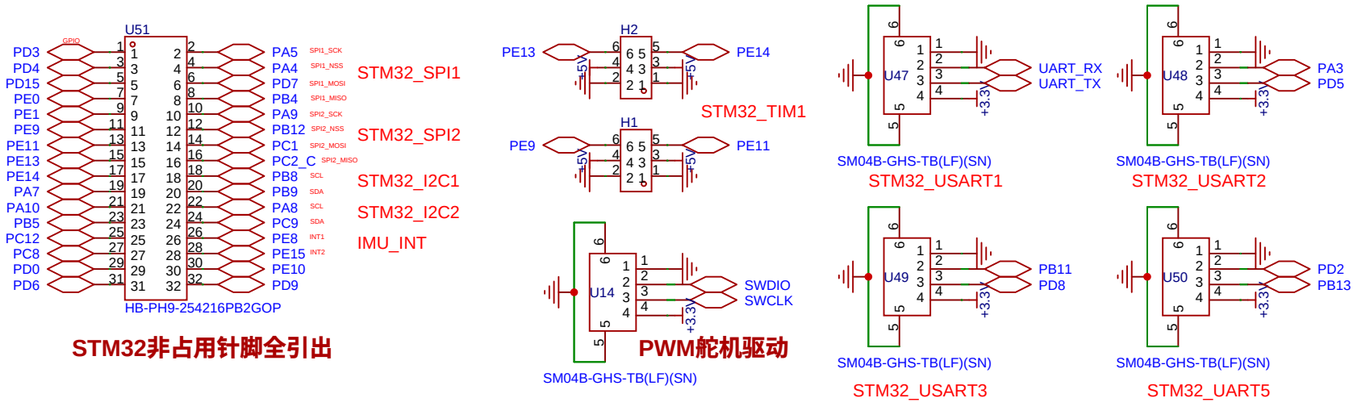


使用双向3.3V_TVS保护芯片，5V TTL会被限压为3.3V

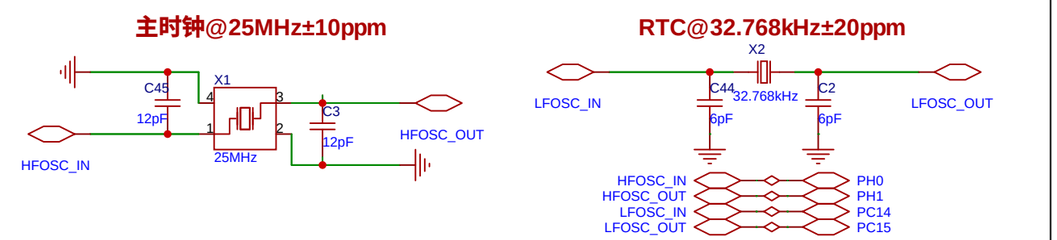
SWD烧录与UART通信模块



声光模块

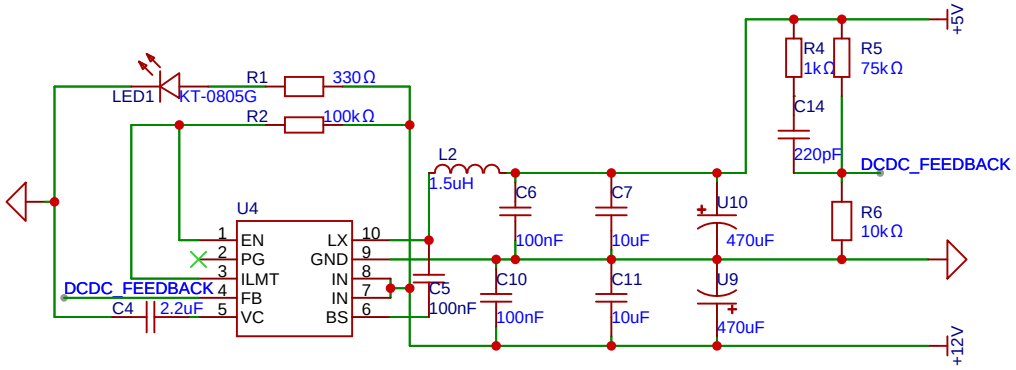


接口及引脚外引模块



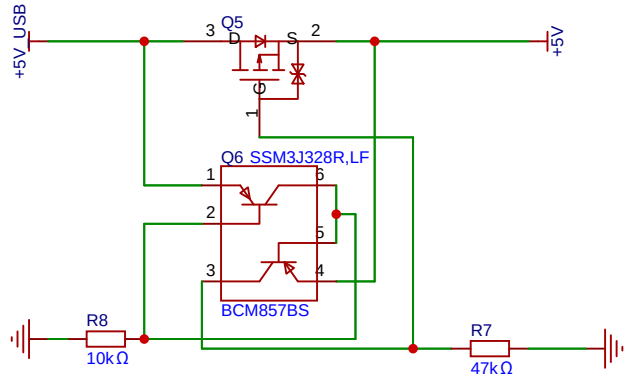
时钟模块

原理图	STM32H750VBT6智能车开发板原理图	创建日期	2025-04-19
		更新日期	2025-04-20
板子	STM32H750VBT6_Car_UniBoard	图页	STM32H750核心功能原理图
绘制	刘振宇	STM32H750VBT6智能车开发板	
审阅	刘鼎琨、宋棋		
		版本	尺寸	页 1 共 3
嘉立创EDA		V2.0	A2	嘉立创EDA



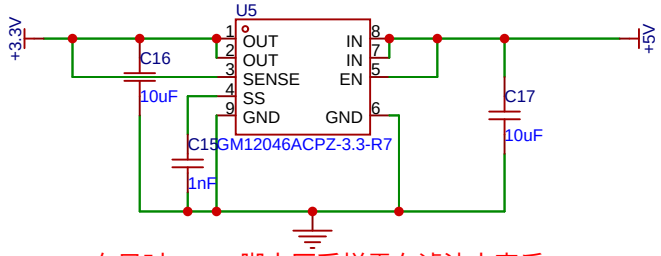
SY8368QNC输入电压范围: 4V-28V

DCDC模块



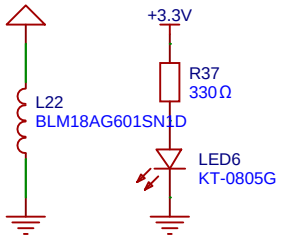
外部电源供电时，断开USB供电，防止灌电流，保护电脑USB口

USB防倒灌保护电路模块



布局时Sense脚电压采样需在滤波电容后

LDO模块

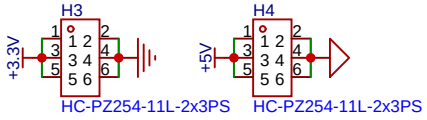


电源指示模块

+12V输入带TVS保护

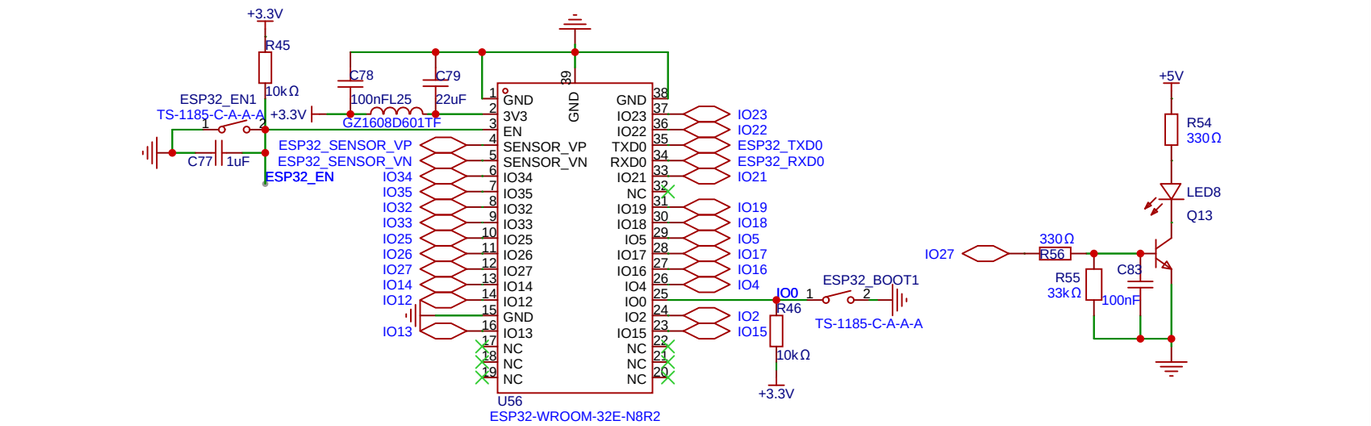
因本模块电流较大，删去磁珠

+12V输入及保护模块

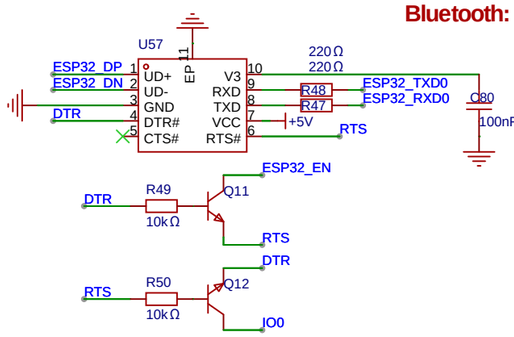


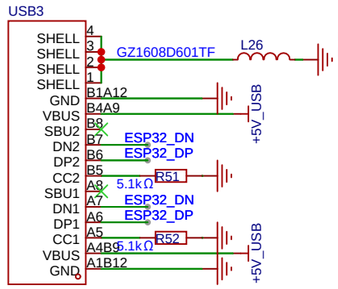
电源引出模块

原理图	STM32H750VBT6智能车开发板原理图	创建日期	2025-04-19
		更新日期	2025-04-20
板子	STM32H750VBT6 Car UniBoard	图页	电源模块原理图
绘制	刘振宇	STM32H750VBT6智能车开发板	
审阅	刘鼎琨、宋棋		
		版本	尺寸	页 2 共 3
嘉立创EDA		V2.0	A2	嘉立创EDA

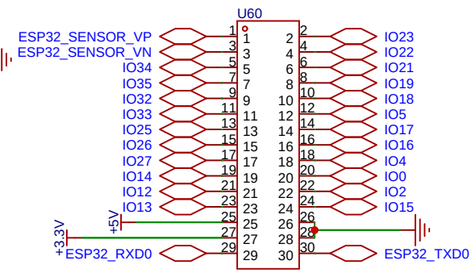


Bluetooth: ESP32-WROOM-32E

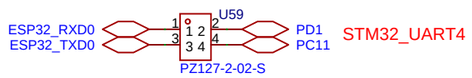




ESP32_USB



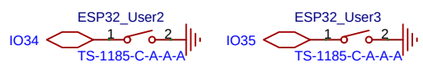
ESP32针脚全引出



STM32_UART4

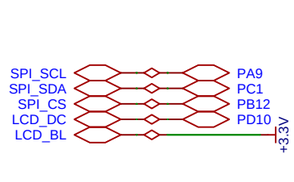
ESP32烧录时，确保与STM32的连接断开

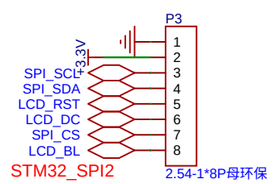
CH340K自动下载电路(ESP32)



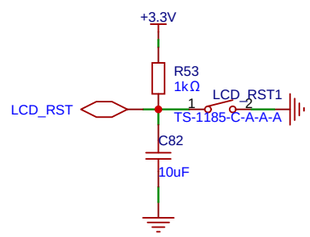
User_Button

蓝牙模块

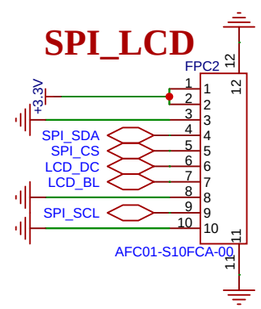




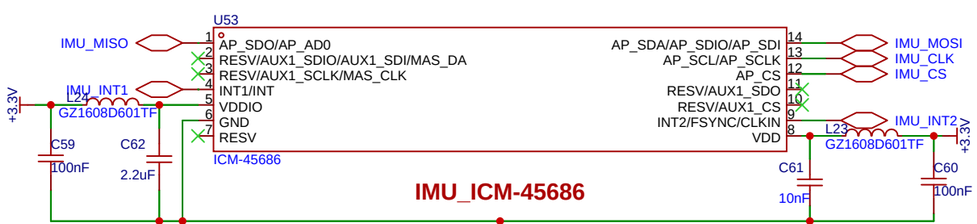
STM32_SPI2

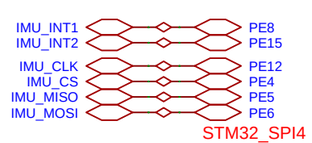


SPI_LCD



LCD模块

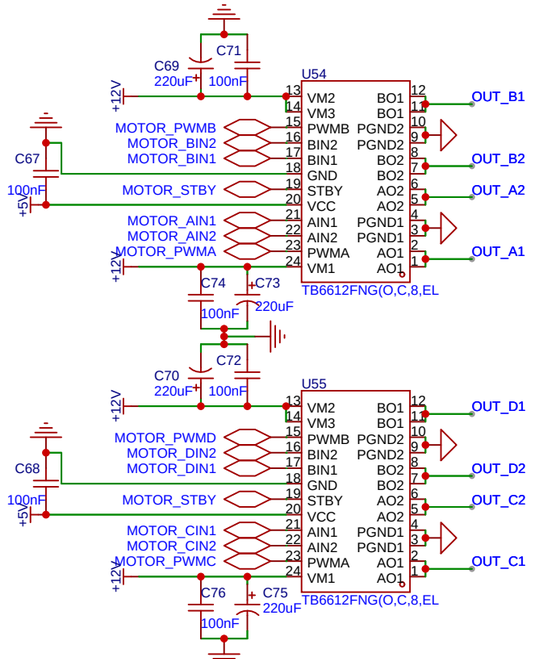


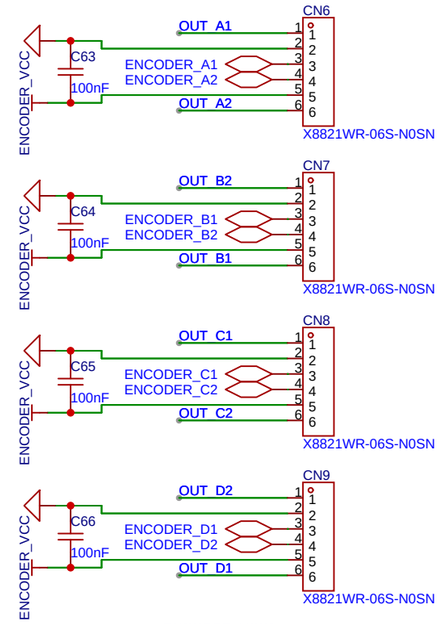


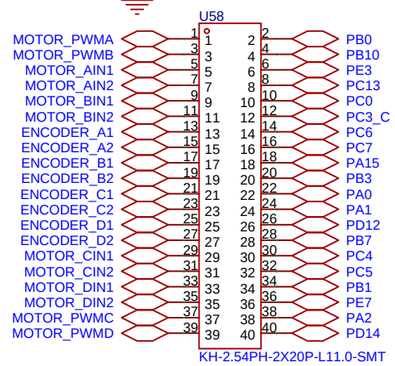
STM32_SPI4

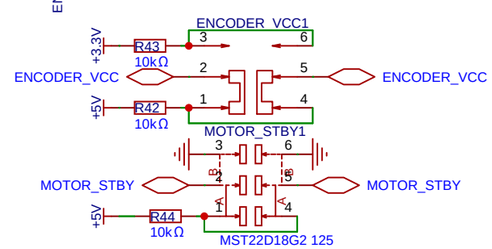
IMU_ICM-45686

IMU模块





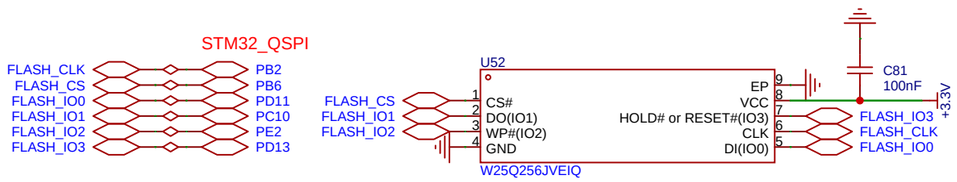




A：左前轮TIM3
B：左后轮（高精Timer）TIM2
C：右后轮（高精Timer）TIM5
D：右前轮TIM4

TB6612驱动模块

STM32_QSPI



FLASH模块

原理图	STM32H750VBT6智能车开发板原理图	创建日期	2025-04-19
		更新日期	2025-04-20
板子	STM32H750VBT6_Car_UniBoard	图页	外设原理图
绘制	刘振宇	STM32H750VBT6智能车开发板	
审阅	刘鼎琨、宋棋		
		版本	尺寸	页 3 共 3
嘉立创EDA		V2.0	A4	嘉立创EDA